

Data Dictionary: Reserve Adequacy Threshold Analysis

Project: Sri Lanka Financial Stress Index (SL-FSI) - Reserve Adequacy Module **Purpose:** Document all raw data sources, column definitions, and derived metrics for the Reserve Level Adequacy Threshold research **Last Updated:** January 2026 **Primary Data Source:** Central Bank of Sri Lanka (CBSL)

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1. Overview

This data dictionary documents all variables used in the Reserve Adequacy Threshold Analysis, which aims to identify ex-ante thresholds that would have signaled Sri Lanka's 2022 sovereign default crisis. The analysis covers multiple reserve adequacy benchmarks including Import Cover, Greenspan-Guidotti Ratio, and IMF ARA Metric.

Key Event Timeline

- April 12, 2022:** Sri Lanka announces sovereign default
- Research Question:** At what reserve level did the crisis become inevitable?
- Data Range:** 2005 - 2025 (with varying coverage by series)

2. Data Source Files

2.1 Primary Source Files (Reserve Adequacy Focus)

File	Source Table	Frequency	Data Range	Description
reserve_assets_monthly_cbsl.csv	CBSL Historical Data Series	Monthly	Nov 2013 - Dec 2025	Official reserve asset components
central_govt_debt_quarterly.csv	CBSL SDDS	Quarterly	2000 - Q3 2025	Central government debt by type
external_debt_usd_quarterly.csv	CBSL Table 2.12	Quarterly	Q4 2012 - Q3 2025	External debt in USD
iip_quarterly_2025.csv	CBSL IIP Release	Quarterly	Q4 2012 - Q3 2025	International Investment Position
monetary_aggregates_monthly.csv	CBSL Table 4.02	Monthly	Dec 1995 - Sep 2025	M0 and M2 money supply
reserve_money_velocity_monthly.csv	CBSL Table 4.11	Monthly	Jan 2003 - Dec 2025	Reserve money and multipliers
monthly_imports_usd.csv	CBSL Table 2.04	Monthly	Jan 2007 - Nov 2025	Merchandise imports
monthly_exports_usd.csv	CBSL Table 2.02	Monthly	Jan 2007 - Nov 2025	Merchandise exports

File	Source Table	Frequency	Data Range	Description
tourism_earnings_monthly.csv	CBSL Table 2.14.1	Monthly	Jan 2009 - Latest	Tourism receipts
remittances_monthly.csv	CBSL Table 2.14.2	Monthly	Jan 2009 - Latest	Workers' remittances
cse_flows_monthly.csv	CBSL Table 2.14.3	Monthly	Jan 2012 - Latest	CSE portfolio flows

2.2 Consolidated Output File

File	Description
slfsl_monthly_panel.csv	Merged monthly panel with all indicators (data/merged/)

3. Column Definitions by Category

3.1 Reserve Assets

Source File: data/external/reserve_assets_monthly_cbsl.csv **CBSL Source:** Official Reserve Assets - Historical Data Series **URL:** <https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector> (<https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector>)

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD
gross_reserves_usd_m	Total official reserve assets	USD millions	Sum of all components below
fx_reserves_usd_m	Foreign currency reserves	USD millions	Primarily held in USD, EUR, GBP
imf_position_usd_m	Reserve position in IMF	USD millions	Sri Lanka's quota contribution
sdrs_usd_m	Special Drawing Rights	USD millions	IMF-allocated SDRs
gold_usd_m	Monetary gold holdings	USD millions	Valued at market prices
other_reserves_usd_m	Other reserve assets	USD millions	Residual category
source	Data provenance marker	String	"CBSL Official Reserve Assets Historical"

Formula:

```
gross_reserves_usd_m = fx_reserves_usd_m + imf_position_usd_m + sdrs_usd_m + gold_usd_m + other_reserves_usd_m
```

3.2 Central Government Debt (LKR)

Source File: data/external/central_govt_debt_quarterly.csv **CBSL Source:** Central Government Debt SDDS (Special Data Dissemination Standard) **URL:** <https://www.cbsl.gov.lk/en/statistics/sdds-sri-lanka> (<https://www.cbsl.gov.lk/en/statistics/sdds-sri-lanka>)

Column	Definition	Unit	Notes
date	Quarter-end date	Date	Format: YYYY-MM-DD
total_debt_lkr_m	Total central government debt	LKR millions	Domestic + Foreign
domestic_debt_lkr_m	Total domestic debt	LKR millions	Rupee-denominated obligations
domestic_short_term_lkr_m	Short-term domestic debt	LKR millions	Maturity < 12 months (T-bills, etc.)
domestic_medium_long_lkr_m	Medium/long-term domestic debt	LKR millions	Maturity ≥ 12 months (T-bonds, etc.)
foreign_debt_lkr_m	Total foreign debt (LKR equivalent)	LKR millions	Converted at prevailing FX rate
foreign_short_term_lkr_m	Short-term foreign debt	LKR millions	Maturity < 12 months
total_short_term_lkr_m	Combined short-term debt	LKR millions	domestic_short_term + foreign_short_term

Column	Definition	Unit	Notes
source	Data provenance marker	String	"CBSL Central Government Debt SDDS"

Formula:

```
total_debt_lkr_m = domestic_debt_lkr_m + foreign_debt_lkr_m
domestic_debt_lkr_m = domestic_short_term_lkr_m + domestic_medium_long_lkr_m
total_short_term_lkr_m = domestic_short_term_lkr_m + foreign_short_term_lkr_m
```

3.3 External Debt (USD)

Source File: data/external/external_debt_usd_quarterly.csv **CBSL Source:** Table 2.12 - Outstanding External Debt **URL:** <https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector> (<https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector>)

Column	Definition	Unit	Notes
date	Quarter-end date	Date	Format: YYYY-MM-DD
govt_total_usd_m	Total government external debt	USD millions	All government foreign obligations
govt_short_term_usd_m	Government short-term external debt	USD millions	Key for Greenspan-Guidotti
govt_long_term_usd_m	Government long-term external debt	USD millions	Maturity ≥ 12 months
central_bank_usd_m	Central bank external debt	USD millions	CBSL foreign obligations
other_sectors_usd_m	Other sectors' external debt	USD millions	SOEs, private sector
total_short_term_usd_m	Combined short-term external debt	USD millions	All sectors, maturity < 12 months

3.4 International Investment Position (IIP)

Source File: data/external/iip_quarterly_2025.csv **CBSL Source:** International Investment Position Release **URL:** <https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector> (<https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector>)

Column	Definition	Unit	Notes
date	Quarter-end date	Date	Format: YYYY-MM-DD
portfolio_equity	Portfolio investment - equity liabilities	USD millions	Foreign holdings of SL equities
portfolio_debt	Portfolio investment - debt securities	USD millions	Foreign holdings of SL bonds
reserve_assets	Official reserve assets (backup)	USD millions	Cross-check with reserve_assets file
total_liabilities	Total external liabilities	USD millions	All foreign claims on SL

Note: For IMF ARA calculation, portfolio liabilities = portfolio_equity + portfolio_debt

3.5 Monetary Aggregates (M0, M2)

Source File: data/external/monetary_aggregates_monthly.csv **CBSL Source:** Table 4.02 - Monetary Aggregates **URL:** <https://www.cbsl.gov.lk/en/statistics/statistical-tables/monetary-sector> (<https://www.cbsl.gov.lk/en/statistics/statistical-tables/monetary-sector>)

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD
reserve_money_m0_lkr_m	Reserve money (M0)	LKR millions	Currency + bank reserves at CBSL
broad_money_m2_lkr_m	Broad money (M2)	LKR millions	Key for IMF ARA calculation

Definition of M2:

```
M2 = Currency in circulation + Demand deposits + Time deposits + Savings deposits
```

3.6 Reserve Money and Multipliers

Source File: data/external/reserve_money_velocity_monthly.csv **CBSL Source:** Table 4.11 - Reserve Money, Multipliers and Velocity of Money **URL:** <https://www.cbsl.gov.lk/en/statistics/statistical-tables/monetary-sector> (<https://www.cbsl.gov.lk/en/statistics/statistical-tables/monetary-sector>)

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD
reserve_money_total_lkr_m	Total reserve money	LKR millions	Monetary base
money_multiplier_m1	M1 money multiplier	Ratio	M1 / Reserve Money

Formula:

```
money_multiplier_m1 = M1 / Reserve Money
```

3.7 Trade Data

Imports

Source File: data/external/monthly_imports_usd.csv **CBSL Source:** Table 2.04 - Monthly Trade Statistics (Imports)

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD
imports_usd_m	Monthly merchandise imports	USD millions	CIF basis

Exports

Source File: data/external/monthly_exports_usd.csv **CBSL Source:** Table 2.02 - Monthly Trade Statistics (Exports)

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD
exports_usd_m	Monthly merchandise exports	USD millions	FOB basis

3.8 External Flows

Tourism Earnings

Source File: data/external/tourism_earnings_monthly.csv **CBSL Source:** Table 2.14.1 - Tourism Earnings

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD
tourism_earnings_usd_m	Monthly tourism receipts	USD millions	Travel services credit

Workers' Remittances

Source File: data/external/remittances_monthly.csv **CBSL Source:** Table 2.14.2 - Workers' Remittances

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD
remittances_usd_m	Monthly remittance inflows	USD millions	Personal transfers from abroad

Colombo Stock Exchange Portfolio Flows

Source File: data/external/cse_flows_monthly.csv **CBSL Source:** Table 2.14.3 - CSE Portfolio Investment Flows

Column	Definition	Unit	Notes
date	Month-end date	Date	Format: YYYY-MM-DD

Column	Definition	Unit	Notes
cse_inflows_usd_m	Foreign portfolio inflows	USD millions	Purchases by non-residents
cse_outflows_usd_m	Foreign portfolio outflows	USD millions	Sales by non-residents
cse_net_usd_m	Net portfolio flows	USD millions	Inflows - Outflows

Formula:

```
cse_net_usd_m = cse_inflows_usd_m - cse_outflows_usd_m
```

4. Derived/Calculated Metrics

4.1 Import Cover Ratio

Location: Merged panel (slfsi_monthly_panel.csv) **Column:** import_cover_months

Formula:

```
Import Cover (months) = Gross Reserves (USD) / Monthly Imports (USD)
```

Thresholds: | Level | Months | Interpretation | |-----|-----|-----| | Comfortable | ≥ 6 | Adequate buffer for external shocks | | IMF Minimum | ≥ 3
| Minimum acceptable level | | Warning | < 2 | Elevated risk of reserves depletion | | Critical | < 1 | Imminent crisis, insufficient cover |

4.2 Net Usable Reserves (PBOC Adjustment)

Location: Merged panel (slfsi_monthly_panel.csv) **Column:** net_usable_reserves_usd_m

Formula:

```
Net Usable Reserves = Gross Reserves - PBOC Swap Amount
```

PBOC Swap Details:

- **Amount:** USD 1,500 million
- **Start Date:** March 1, 2021
- **Status:** Encumbered (subject to conditionalities, not freely usable)

Rationale: The China-Sri Lanka bilateral currency swap was activated in March 2021 but is subject to usage conditions and must be repaid. It inflates gross reserves without providing equivalent liquidity.

4.3 Net Import Cover

Location: Merged panel (slfsi_monthly_panel.csv) **Column:** net_import_cover_months

Formula:

```
Net Import Cover (months) = Net Usable Reserves (USD) / Monthly Imports (USD)
```

4.4 Greenspan-Guidotti Ratio

Calculated in: app_reserve_adequacy.py

Formula:

$GG\ Ratio = \frac{Gross\ Reserves\ (USD)}{Short-term\ External\ Debt\ (USD)}$

Data Sources:

- Numerator: `gross_reserves_usd_m` from reserve assets
- Denominator: `govt_short_term_usd_m` from external debt (quarterly)

Threshold: GG $\geq$ 1.0 indicates ability to service all short-term debt without external financing.

Conversion for LKR-denominated debt:

$Short-term\ Debt\ (USD) = \frac{total_short_term_lkr_m}{usd_lkr}$

4.5 IMF Assessing Reserve Adequacy (ARA) Metric

Calculated in: `app_reserve_adequacy.py`

Formula:

$ARA = 5\% \times Annual\ Exports + 5\% \times Broad\ Money\ (M2\ in\ USD) + 30\% \times Short-term\ Debt + 15\% \times Portfolio\ Liabilities$

Component Breakdown:

Component	Weight	Source Column	Transformation
Annual Exports	5%	<code>exports_usd_m</code>	Sum quarterly, annualize ($\times 4$)
Broad Money M2	5%	<code>broad_money_m2_lkr_m</code>	Convert to USD using <code>usd_lkr</code>
Short-term Debt	30%	<code>govt_short_term_usd_m</code>	Direct use (quarterly)
Portfolio Liabilities	15%	<code>portfolio_equity + portfolio_debt</code>	Sum from IIP (quarterly)

ARA Ratio:

$ARA\ Ratio = \frac{Actual\ Reserves}{ARA\ Requirement}$

Threshold: ARA Ratio $\geq$ 100% indicates adequate reserves; 100-150% is the recommended range.

4.6 Real Policy Rate

Location: Merged panel (`slfsi_monthly_panel.csv`) **Column:** `real_policy_rate`

Formula:

$Real\ Policy\ Rate = Nominal\ Policy\ Rate - Inflation\ (YoY)$

Data Sources:

- `policy_ceiling` (CBSL Standing Lending Facility Rate)
- `ncpi_yoy_pct` (National CPI, year-on-year percentage change)

5. Data Coverage Summary

5.1 Coverage by Series (Crisis Period: 2020-2024)

Metric	Start Date	End Date	Frequency	Crisis Coverage
Gross Reserves	Nov 2013	Dec 2025	Monthly	☐ Complete
Government Debt (LKR)	Dec 2000	Q3 2025	Quarterly	☐ Complete
External Debt (USD)	Q4 2012	Q3 2025	Quarterly	☐ Complete
Portfolio Liabilities	Q4 2012	Q3 2025	Quarterly	☐ Complete
Broad Money M2	Dec 1995	Sep 2025	Monthly	☐ Complete
Imports	Jan 2007	Nov 2025	Monthly	☐ Complete
Exports	Jan 2007	Nov 2025	Monthly	☐ Complete
Tourism	Jan 2009	Nov 2025	Monthly	☐ Complete
Remittances	Jan 2009	Nov 2025	Monthly	☐ Complete
CSE Flows	Jan 2012	Dec 2025	Monthly	☐ Complete
FX Rate (USD/LKR)	Jan 2005	Dec 2025	Daily/Monthly	☐ Complete

5.2 Data Limitations

- Quarterly vs Monthly:** Debt data is quarterly, requiring interpolation or quarterly-only analysis
- PBOC Swap Opacity:** Exact terms and utilization of the swap are not publicly disclosed
- Single Crisis Sample:** Cannot fully backtest against 2008-2009 or 2011-2012 stress periods due to data gaps
- Reporting Lags:** Official CBSL data published with 1-2 month lag

6. References and URLs

6.1 Primary Data Sources

Source	URL
CBSL Statistical Tables	https://www.cbsl.gov.lk/en/statistics/statistical-tables (https://www.cbsl.gov.lk/en/statistics/statistical-tables)
CBSL External Sector Statistics	https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector (https://www.cbsl.gov.lk/en/statistics/statistical-tables/external-sector)
CBSL Monetary Sector Statistics	https://www.cbsl.gov.lk/en/statistics/statistical-tables/monetary-sector (https://www.cbsl.gov.lk/en/statistics/statistical-tables/monetary-sector)
CBSL SDDS (Debt Data)	https://www.cbsl.gov.lk/en/statistics/sdds-sri-lanka (https://www.cbsl.gov.lk/en/statistics/sdds-sri-lanka)
CBSL Data Library	https://www.cbsl.gov.lk/en/statistics/statistical-data-library (https://www.cbsl.gov.lk/en/statistics/statistical-data-library)

6.2 Specific CBSL Tables Referenced

Table	Description	Data Category
Table 2.02	Monthly Exports	Trade
Table 2.04	Monthly Imports	Trade
Table 2.12	Outstanding External Debt	External Debt
Table 2.14.1	Tourism Earnings	External Flows
Table 2.14.2	Workers' Remittances	External Flows
Table 2.14.3	CSE Portfolio Flows	External Flows
Table 4.02	Monetary Aggregates	Money Supply
Table 4.11	Reserve Money, Multipliers, Velocity	Money Supply
IIP Release	International Investment Position	External Position
SDDS	Central Government Debt	Government Debt
Historical Series	Official Reserve Assets	Reserves

6.3 Methodology References

Metric

Greenspan-Guidotti Rule Greenspan, A. (1999). "Currency reserves and debt." IMF/World Bank remarks
IMF ARA Metric IMF (2011). "Assessing Reserve Adequacy." IMF Policy Paper
Import Cover IMF traditional metric; 3-month minimum guideline

Reference

Appendix: Column Header Cross-Reference

This table maps the column headers specified in the research plan to their actual file locations:

Research Plan Header	Actual Column	File Location
total_debt_lkr_m	total_debt_lkr_m	central_govt_debt_quarterly.csv
domestic_debt_lkr_m	domestic_debt_lkr_m	central_govt_debt_quarterly.csv
domestic_short_term_lkr_m	domestic_short_term_lkr_m	central_govt_debt_quarterly.csv
domestic_medium_long_lkr_m	domestic_medium_long_lkr_m	central_govt_debt_quarterly.csv
foreign_debt_lkr_m	foreign_debt_lkr_m	central_govt_debt_quarterly.csv
foreign_short_term_lkr_m	foreign_short_term_lkr_m	central_govt_debt_quarterly.csv
total_short_term_lkr_m	total_short_term_lkr_m	central_govt_debt_quarterly.csv
portfolio_liabilities	portfolio_equity + portfolio_debt	iip_quarterly_2025.csv (derived)
gross_reserves	gross_reserves_usd_m	reserve_assets_monthly_cbsl.csv
import_cover_months	import_cover_months	slfsi_monthly_panel.csv (derived)
Reserve_money_total_lkr	reserve_money_total_lkr_m	reserve_money_velocity_monthly.csv
money_multiplier_m1	money_multiplier_m1	reserve_money_velocity_monthly.csv
cse_inflows_usd_m	cse_inflows_usd_m	cse_flows_monthly.csv
cse_outflows_usd_m	cse_outflows_usd_m	cse_flows_monthly.csv
cse_net_usd_m	cse_net_usd_m	cse_flows_monthly.csv
remittances_usd_m	remittances_usd_m	remittances_monthly.csv
tourism_earnings_usd_m	tourism_earnings_usd_m	tourism_earnings_monthly.csv
reserve_money_m0_lkr_m	reserve_money_m0_lkr_m	monetary_aggregates_monthly.csv
broad_money_m2_lkr_m	broad_money_m2_lkr_m	monetary_aggregates_monthly.csv
gross_reserves_usd_m	gross_reserves_usd_m	reserve_assets_monthly_cbsl.csv
fx_reserves_usd_m	fx_reserves_usd_m	reserve_assets_monthly_cbsl.csv
imf_position_usd_m	imf_position_usd_m	reserve_assets_monthly_cbsl.csv
sdrs_usd_m	sdrs_usd_m	reserve_assets_monthly_cbsl.csv
gold_usd_m	gold_usd_m	reserve_assets_monthly_cbsl.csv
other_reserves_usd_m	other_reserves_usd_m	reserve_assets_monthly_cbsl.csv
imports_usd_m	imports_usd_m	monthly_imports_usd.csv
import_cover	import_cover_months	slfsi_monthly_panel.csv (derived)
net_reserves	net_usable_reserves_usd_m	slfsi_monthly_panel.csv (derived)
net_import_cover	net_import_cover_months	slfsi_monthly_panel.csv (derived)
govt_total_usd_m	govt_total_usd_m	external_debt_usd_quarterly.csv